26-BS-M201 — Mathematics – II

Course Code	26-BS-M201	**Category**	Basic Science Course
Course Title	Mathematics – II	**L – T – P**	3 – 0 – 0
Credits	3	**Contact Hours**	36
Semester	Second	**Assessment**	Theory: CA 30 + ESE 70

Prerequisites

Continues Mathematics-I and, like it, is generic to every branch. Assumed:

- The differential and integral calculus of Mathematics-I.
- Elementary vectors and the dot and cross product (CBSE XII Unit IV).
- First-order differential equations by separation of variables (CBSE XII Unit IX).
- Complex numbers in algebraic and polar form (CBSE XI; WBCHSE).

Course Objectives

- To develop the solution of ordinary differential equations, the principal analytical tool of engineering.
- To establish the transform methods and the vector calculus used across the later curriculum.
- To introduce the elements of complex-variable and Fourier methods on a common footing.
- To keep the treatment branch-neutral, discipline-specific mathematics being reserved to Mathematics-III.

Course Outcomes

On completion of the course the student will be able to:

CO	Statement	Bloom
CO1	Solve linear ordinary differential equations of first and higher order and apply them to elementary engineering models.	L3 – Apply
CO2	Apply the Laplace transform and its inverse to the solution of differential equations and initial-value problems.	L3 – Apply
CO3	Apply the differential and integral calculus of vectors, including the theorems of Green, Stokes and Gauss in outline.	L3 – Apply
CO4	Use the elements of complex-variable theory and of Fourier series in the analysis of engineering functions.	L4 – Analyse

CO–PO Articulation Matrix

Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	1	1	–	–	–	–	–	1
CO2	3	2	2	1	2	–	–	–	–	–	1
CO3	3	2	1	1	1	–	–	–	–	–	1
CO4	3	3	1	1	1	–	–	–	–	–	2

Course Plan — Module-wise Syllabus

Module	Title	Contents	Hours
I	Ordinary Differential Equations	Linear differential equations of the first order; exact equations and integrating factors; linear equations of higher order with constant coefficients; complementary function and particular integral; variation of parameters in outline; the Cauchy–Euler equation; elementary applications to growth, decay and oscillation, drawn generically.	9
II	Laplace Transform	The Laplace transform and its properties; transforms of standard functions; the inverse transform; the shifting theorems; the transform of derivatives and integrals; solution of linear differential equations and initial-value problems by the transform method; the unit-step function in outline.	9
III	Vector Calculus	Scalar and vector fields; gradient, divergence and curl and their physical meaning; line, surface and volume integrals; the theorems of Green, Gauss and Stokes in statement and elementary use.	9
IV	Complex Variables and Fourier Series	Functions of a complex variable; analyticity and the Cauchy–Riemann equations; harmonic functions; Cauchy's integral theorem and formula (statement, elementary use). Periodic functions and their Fourier-series representation; the Euler formulae; even and odd functions and half-range series. The Fourier and partial-differential extensions are left to Mathematics-III.	9

Cross-cutting notes. This is a generic mathematics course, common to every branch of the University; it deliberately carries no discipline-specific material. Where a programme requires mathematics particular to it — discrete structures and graph theory for the computing branches, additional transforms and complex analysis for the electronics and electrical branches, additional statistics and numerical methods for the others — that requirement is to be met either in Mathematics-III in Semester III, which each Department frames for its own programme, or in Module I of the relevant discipline course where it is small enough to be carried there. The first-year mathematics is thus kept uniform, so that the common first-year examination and inter-branch mobility are preserved.

Text and Reference Books

- Reena Garg, Mathematics (Calculus and Linear Algebra), Khanna Publishing House (AICTE Prescribed Textbook)
- Reena Garg, Advanced Engineering Mathematics, Khanna Publishing House.
- Erwin Kreyszig, Advanced Engineering Mathematics, Wiley.
- M. D. Raisinghania, Ordinary and Partial Differential Equations, S. Chand.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
- N. P. Bali and M. Goyal, A Textbook of Engineering Mathematics, Laxmi Publications.

Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Differential Equations for Engineers, 12 weeks.
- NPTEL — Integral and Vector Calculus, 12 weeks.
- NPTEL — Advanced Engineering Mathematics, 12 weeks.
- SWAYAM — Mathematics II (AICTE / SWAYAM), as offered.

Question-paper note. As in Mathematics-I the paper is predominantly a problem paper, and no question draws on discipline-specific material. At least one Group C question requires the solution of a differential equation by two distinct methods.